

FORT WORTH ALLIANCE, TX, USA

WMO: 722594

Lat: 32.973N

Lon: 97.318W

Elev: 209

StdP: 98.84

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-3.7	-13.0	1.3	-1.4	-10.9	1.5	1.5	12.1	9.2	10.9	10.3	3.3	0	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	39.0	23.3	37.7	23.6	36.4	23.5	25.7	33.6	25.3	33.1	24.9	32.5	4.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.1	28.8	23.0	18.2	28.0	22.7	17.9	27.9	80.4	33.9	78.2	33.2	76.4	32.4	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.5	9.2	8.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-9.7	40.9	2.1	1.9	-11.3	42.3	-12.5	43.4	-13.7	44.4	-15.3	45.8
			-10.5	26.7	2.0	0.8	-11.9	27.3	-13.1	27.7	-14.2	28.1	-15.7	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	7.7	9.8	14.1	18.3	23.0	27.7	29.8	29.9	25.6	19.5	13.5	8.3
	DBStd	9.10	5.62	5.87	5.43	4.32	3.74	2.60	2.40	2.79	3.69	4.77	5.27	5.36
	HDD10.0	331	112	69	23	2	0	0	0	0	0	2	25	97
	HDD18.3	1306	332	244	150	52	8	0	0	0	2	44	161	313
	CDD10.0	3609	41	63	151	252	404	530	612	616	469	297	129	44
	CDD18.3	1541	2	4	19	53	154	280	354	357	221	81	14	2
	CDH23.3	18603	15	41	158	435	1483	3388	4786	4876	2518	784	103	16
	CDH26.7	9580	1	6	30	99	555	1734	2749	2867	1243	276	18	2
(14) Wind	WSAvg	3.8	4.0	4.1	4.5	4.6	4.4	4.0	3.5	3.1	3.0	3.5	3.7	3.7
(15) Precipitation	PrecAvg	897	54	54	73	95	113	85	55	58	74	102	74	61
	PrecMax	1443	144	179	157	197	343	314	127	184	266	275	242	252
	PrecMin	369	4	0	0	16	12	10	0	0	16	12	2	9
	PrecStd	243	34	46	44	45	68	71	39	45	55	72	61	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	27.4	29.6	31.9	35.5	38.3	40.7	41.3	38.8	33.9	28.8	25.9
		MCWB	14.6	15.7	18.5	20.9	23.1	23.4	23.1	23.1	22.7	21.7	18.8	16.7
	2%	DB	22.4	24.1	27.1	29.0	33.3	36.7	38.8	39.5	36.2	31.8	26.2	22.5
		MCWB	14.4	15.8	18.2	19.9	22.7	23.7	23.4	23.3	22.9	21.3	18.0	15.2
	5%	DB	19.8	22.1	25.0	27.5	31.6	35.2	37.5	37.9	34.4	29.7	23.9	20.1
		MCWB	13.4	15.4	17.0	19.6	22.3	23.5	24.0	23.5	22.6	20.3	17.4	14.6
	10%	DB	17.5	19.6	22.9	26.0	29.9	33.8	36.1	36.8	32.8	27.8	22.1	17.6
		MCWB	12.6	14.1	16.3	18.8	21.8	23.5	23.8	23.5	22.4	19.5	16.4	12.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	19.6	21.0	23.3	25.6	26.6	26.4	26.0	25.5	24.0	21.6	19.2
		MCDB	21.5	23.8	26.8	28.3	31.8	34.0	34.9	34.2	32.8	29.3	25.4	22.6
	2%	WB	17.0	18.5	19.7	22.0	24.5	25.6	25.7	25.4	24.7	23.0	20.0	17.7
		MCDB	19.7	21.7	24.5	26.6	30.0	32.7	34.0	34.0	31.7	28.4	23.6	21.2
	5%	WB	15.1	16.9	18.8	21.1	23.7	25.0	25.2	25.0	24.1	22.3	18.8	15.5
		MCDB	18.6	20.2	23.2	25.5	29.1	31.9	33.4	33.5	30.8	27.3	22.5	18.5
	10%	WB	12.6	15.0	17.7	20.2	22.9	24.4	24.8	24.5	23.5	21.3	17.5	13.4
		MCDB	16.9	18.9	21.8	24.4	28.2	31.1	32.8	32.7	30.0	25.8	21.2	17.2
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	12.1	12.2	11.9	11.0	11.1	11.6	12.0	12.1	12.5	12.0	11.7
		MCDBR	16.1	15.3	14.3	13.2	12.5	12.5	13.1	13.4	13.4	14.0	13.6	14.9
	5% WB	MCWBR	9.6	9.1	7.6	6.9	5.1	3.7	3.5	3.5	4.1	6.2	7.5	9.0
		MCWBR	12.9	12.9	11.5	10.4	10.5	10.5	11.3	11.5	10.9	10.9	11.3	11.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.311	0.323	0.355	0.390	0.423	0.435	0.437	0.436	0.398	0.358	0.334	0.316
		taud	2.499	2.473	2.403	2.331	2.299	2.301	2.333	2.320	2.373	2.470	2.476	2.485
	Ebn at Noon	Ebn at Noon	909	932	922	899	868	853	849	846	867	881	873	881
		Edh at Noon	86	97	113	127	132	132	127	127	115	97	87	83
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.91	3.44	4.51	5.47	5.89	6.63	6.74	6.18	5.17	4.05	3.08	2.55
	RadStd	RadStd	0.29	0.54	0.34	0.45	0.58	0.63	0.46	0.39	0.45	0.60	0.40	0.32

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page